

Workshop 8 TEMPLATE

12 PM

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2026-04-22

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In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

Specific leaf area is a continuous variable measured in cm^2/g .

Water treatment is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

Species is a categorical variable (again, a *factor*) with 6 levels.

Species name	Species code	Common name
<i>Encelia californica</i>	ENCCAL	Bush sunflower
<i>Eschscholzia californica</i>	ESCCAL	California poppy
<i>Penstemon centranthifolius</i>	PENCEN	Scarlet bugler
<i>Grindelia camporum</i>	GRINCAM	Gumweed
<i>Salvia leucophylla</i>	SALLEU	Purple sage
<i>Stipa pulchra</i>	STIPUL	Purple needlegrass
<i>Lotus scoparius</i>	LOTSCO	Deerweed

1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(MuMin) # model selection

# read in your data here
drought_exp <- read_xlsx(here("data", "Valliere_etal_EcoApps_Data.xlsx"), sheet = "First Harvest")

# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
salleu_col <- "#6D397D"
enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
esccal_col <- "#515481"
gricam_col <- "#6C91BD"

# storing colors to use for water treatments
```

```
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw())
```

2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```
# creating new object
drought_exp_clean <- drought_exp |>
  # cleaning column names - lowercase with no spaces (_)
  clean_names() |>
  # renaming species names
  mutate(species_name = recode_values(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
    "STIPUL" ~ "Stipa pulchra", # purple needlegrass
    "LOTSCO" ~ "Lotus scoparius" # deerweed
  )) |>
  # renaming water treatments
  mutate(water_treatment = case_match(
    water,
    "WW" ~ "Well watered",
    "DS" ~ "Drought stressed"
  )) |>
  # assigning order of variables
  mutate(species_name = as_factor(species_name),
    species_name = fct_relevel(species_name,
      "Lotus scoparius",
      "Penstemon centranthifolius",
      "Salvia leucophylla",
      "Encelia californica",
      "Stipa pulchra",
```

```

                                "Eschscholzia californica",
                                "Grindelia camporum")) |>
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                     "Drought stressed",
                                     "Well watered")) |>

# selecting specific columns of interest
select(species_name, water_treatment, sla, total_g)

```

Insert code to double check the structure of the data frame here.

```
str(drought_exp_clean)
```

```

tibble [70 x 4] (S3: tbl_df/tbl/data.frame)
 $ species_name   : Factor w/ 7 levels "Lotus scoparius",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ water_treatment: Factor w/ 2 levels "Drought stressed",...: 2 2 2 2 2 1 1 1 1 1 ...
 $ sla            : num [1:70] 170 215 209 216 222 ...
 $ total_g        : num [1:70] 0.455 0.329 0.45 0.359 0.352 ...

```

Insert code to display 10 random rows from the data frame here.

```

# clean data frame displaying 10 rows
slice_sample(drought_exp_clean, n = 10)

```

```

# A tibble: 10 x 4
  species_name      water_treatment    sla total_g
  <fct>            <fct>          <dbl>  <dbl>
1 Eschscholzia californica Drought stressed 109.   0.298
2 Eschscholzia californica Well watered     117.   0.574
3 Stipa pulchra      Drought stressed 140.   0.359
4 Lotus scoparius    Well watered     123.   0.182
5 Eschscholzia californica Drought stressed 118.   0.129
6 Penstemon centranthifolius Drought stressed 175.   0.188
7 Salvia leucophylla  Well watered     113.   0.195
8 Grindelia camporum  Drought stressed 167.   0.314
9 Grindelia camporum  Well watered     213.   0.678
10 Lotus scoparius    Drought stressed  91.4  0.0626

```

3. Visualizing data

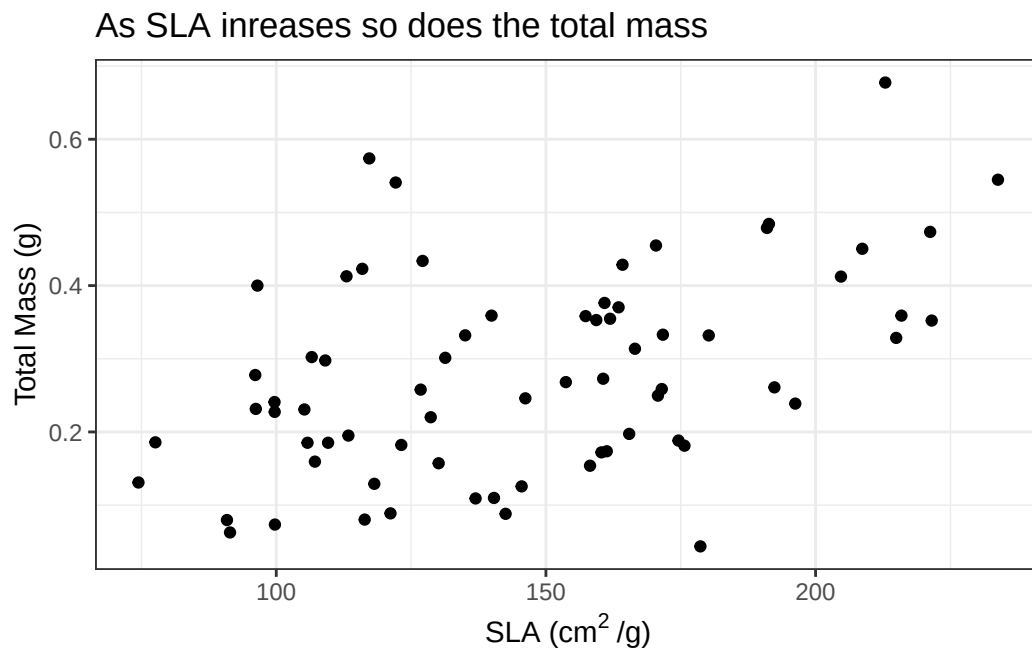
What is the relationship between SLA and plant mass?

Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass _____).

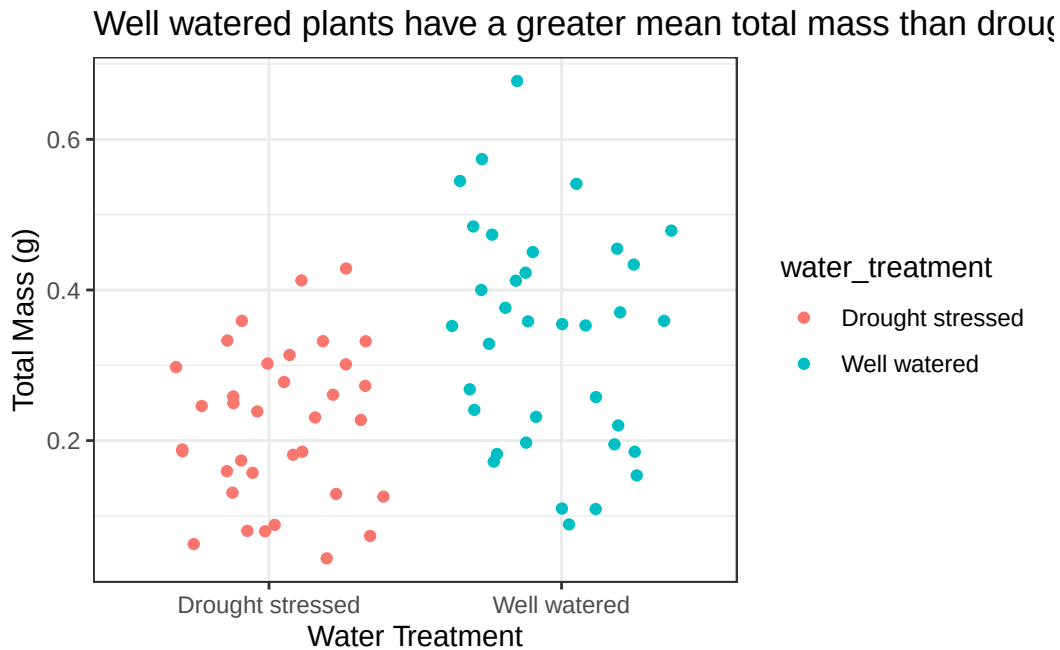
```
ggplot(data = drought_exp_clean, mapping = aes(x = sla, y = total_g)) +  
  geom_point() +  
  labs(x = "SLA (cm2/g)",  
       y = "Total Mass (g)",  
       title = "As SLA increases so does the total mass")
```



What are the differences in total mass between water treatments?

Make a jitterplot to visualize the differences in plant mass between water treatments.

```
ggplot(data = drought_exp_clean, mapping = aes(x = water_treatment,
                                                y = total_g,
                                                color = water_treatment)) +
  geom_jitter(height = 0) +
  labs(x = "Water Treatment", y = "Total Mass (g)", title = "Well watered plants have a greater mean total mass than drought stressed plants")
```



Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between water treatments.

What are the differences in total mass between species?

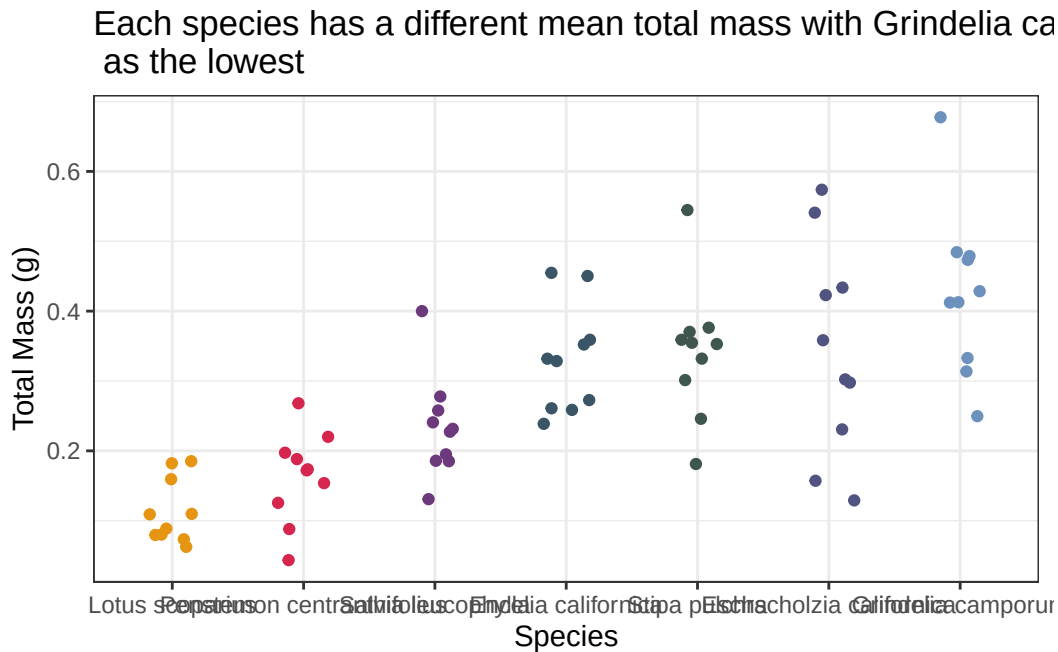
Make a jitterplot to visualize the differences in plant mass between species.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

```
ggplot(data = drought_exp_clean, mapping = aes(x = species_name,
                                                y = total_g,
                                                color = species_name)) +
```

```
geom_jitter(height = 0, width = 0.2) +
  labs( x = "Species", y = "Total Mass (g)", title = "Each species has a different mean total mass, with Salvia leucophylla
as the lowest") +
  theme(legend.position = "none") +
  scale_color_manual(values = c("Lotus scoparius" = lotsco_col,
                                "Penstemon centranthifolius" = pencen_col, "Salvia leucophylla" = salvia_col))
```



Bonus: order the x-axis by decreasing mean plant mass.

4. Fitting models

8 models total:

Model number	SLA	Water treatment	Species	Predictor list
0				no predictors (null model)
1	X	X	X	all predictors (full model)
2	X	X		SLA and water treatment
3	X		X	SLA and species
4		X	X	water treatment and species
5	X			SLA
6		X		water treatment

Model number	SLA	Water treatment	Species	Predictor list
7			X	species

Model fitting

```
# model 0: null model
model0 <- lm(total_g ~ 1, data = drought_exp_clean)

# model 1: all predictors
model1 <- lm(total_g ~ water_treatment + species_name + sla, data = drought_exp_clean)

# model 2: SLA and water treatment
model2 <- lm(total_g ~ sla + water_treatment, data = drought_exp_clean)

# model 3: SLA and species
model3 <- lm(total_g ~ sla + species_name, data = drought_exp_clean)

# model 4: water treatment and species
model4 <- lm(total_g ~ water_treatment + species_name, data = drought_exp_clean)

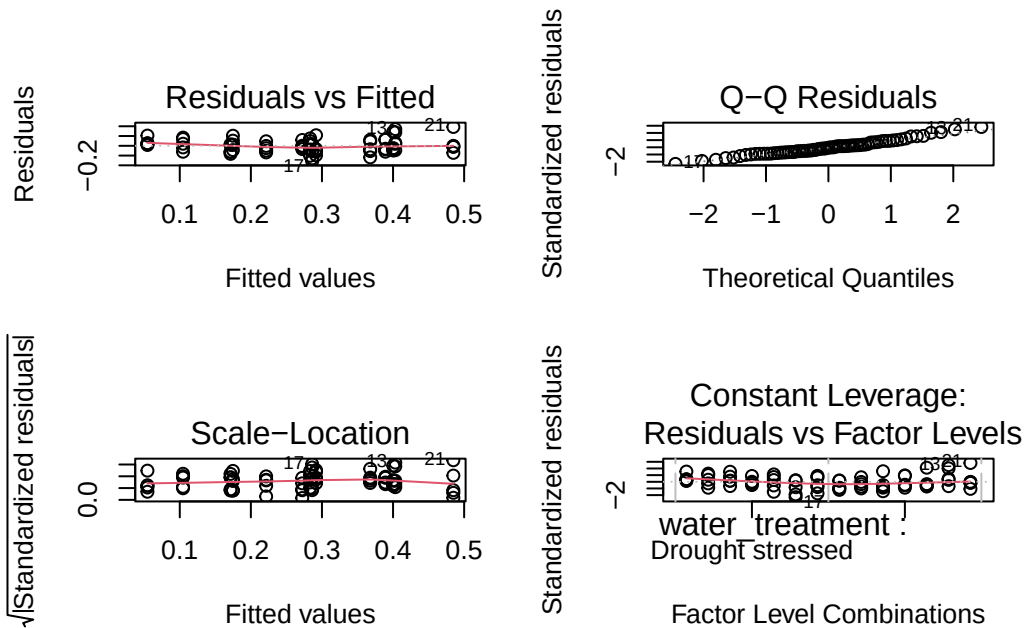
# model 5: SLA
model5 <- lm(total_g ~ sla, data = drought_exp_clean)

# model 6: water treatment
model6 <- lm(total_g ~ water_treatment, data = drought_exp_clean)

# model 7: species
model7 <- lm(total_g ~ species_name, data = drought_exp_clean)
```

Model diagnostics

```
par(mfrow = c(2,2))
plot(model4)
```

Residuals are normally distributed from the qqplot residuals are homoscedastic - residuals vs fitted/scale location no outliers that could be influencing line of best fit

Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
AICc(model10, model11, model12, model13, model14, model15, model16, model17) |>
  arrange(AICc)
```

	df	AICc
model14	9	-156.19595
model11	10	-153.75361
model13	9	-124.07569
model17	8	-120.30191
model12	4	-95.82521
model15	3	-88.85180
model16	3	-86.76661
model10	2	-74.98036

Insert code to display the summary for the best model.

```
summary(model4)
```

Call:

```
lm(formula = total_g ~ water_treatment + species_name, data = drought_exp_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.157087	-0.046953	-0.003733	0.041244	0.192657

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.05455	0.02451	2.225	0.02973 *
water_treatmentWell watered	0.11695	0.01733	6.746	5.90e-09 ***
species_namePenstemon centranthifolius	0.05003	0.03243	1.543	0.12799
species_nameSalvia leucophylla	0.12020	0.03243	3.706	0.00045 ***
species_nameEncelia californica	0.21774	0.03243	6.714	6.70e-09 ***
species_nameStipa pulchra	0.22881	0.03243	7.055	1.72e-09 ***
species_nameEschscholzia californica	0.23164	0.03243	7.143	1.22e-09 ***
species_nameGrindelia camporum	0.31335	0.03243	9.662	5.53e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07252 on 62 degrees of freedom

Multiple R-squared: 0.7535, Adjusted R-squared: 0.7257

F-statistic: 27.08 on 7 and 62 DF, p-value: < 2.2e-16

Stop and think: what does this model mean?

What is the best model?

The best model is model 4 - which is water treatment and species as predictors.

How much variation in the response (total mass, in grams) does this model explain?

72.6% of the variation in plant mass is explained by this model.

How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

Species influences plant mass. Treatment influences plant mass. - The (intercept) is the model prediction for plant mass for *Lotus scoparius* in the drought stressed treatment. - The `water_treatmentWell watered`: model prediction for the difference in *Lotus scoparius* mass between well watered and drought stressed treatments. - `species_namePenstemon centranthifolius`: difference between *Penstemon* mass in well watered treatments and the *Lotus scoparius* mass in the well watered treatment. - `species_nameEschscholzia californica`: difference between the *Eschscholzia californica* mass in well watered treatment and the *Lotus scoparius* mass in the well watered treatment.

4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

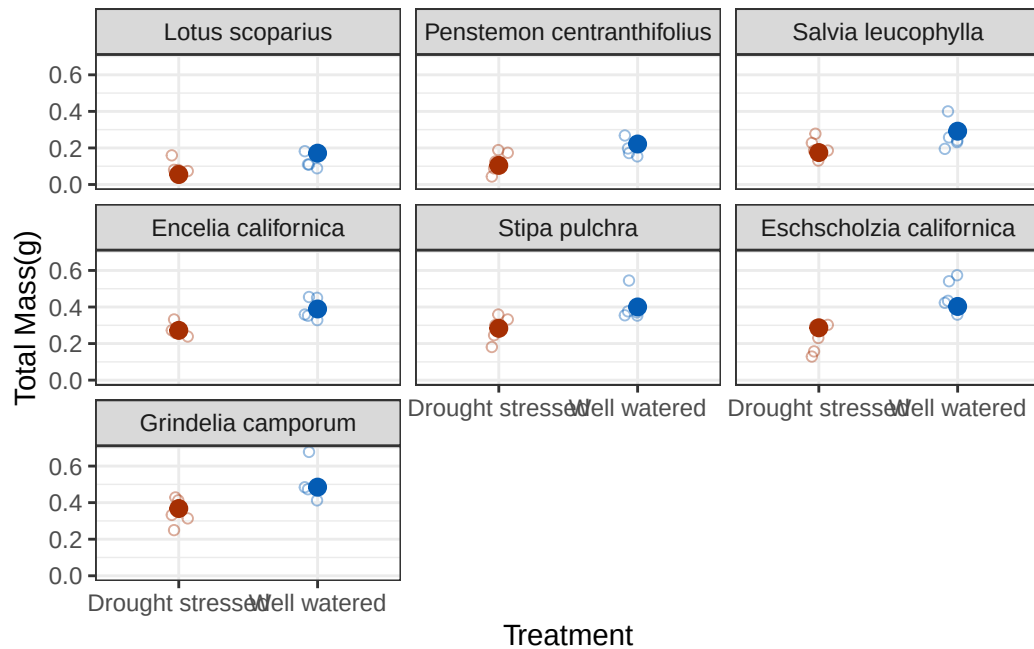
Look at the object before plotting.

```
preds <- ggpredict(model4, terms = c("water_treatment", "species_name")) |> # predictors and
rename(water_treatment = x, species_name = group
)
```

5. Final figure

Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```
ggplot(data = drought_exp_clean, mapping = aes(x = water_treatment, y = total_g, color = water_treatment)) +
  geom_jitter(height = 0, width = 0.1, alpha = 0.4, shape = 21) +
  geom_pointrange(data = preds, mapping = aes(x = water_treatment, y = predicted, ymin = conf_low, ymax = conf_high)) +
  facet_wrap(~ species_name) +
  scale_color_manual(values = c("Drought stressed" = ds_col, "Well watered" = ww_col)) +
  theme(legend.position = "none") + labs(x = "Treatment", y = "Total Mass(g)")
```



6. Writing

Figure caption

Basic components of test

predictors

response

test

distribution

parameters (degrees of freedom)

R^2

p-value

significance level

reference level(s)

Written communication

Methods

We used linear models to determine the predictive relationship between specific leaf area (cm^2/g), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

Results

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model: $F(2, 67) = 14.57$, $p < 0.001$, $\alpha = 0.05$).